



Distinct mechanisms mediate pacemaker dysfunction associated with catecholaminergic polymorphic ventricular tachycardia mutations: Insights from computational modeling



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ABSTRACT

Catecholaminergic polymorphic ventricular tachycardia (CPVT) is a stress-induced ventricular arrhythmia associated with rhythm disturbance and impaired sinoatrial node cell (SANC) automaticity (pauses). Mutations associated with dysfunction of Ca^{2+} -related mechanisms have been shown to be present in CPVT. These dysfunctions include impaired Ca^{2+} release from the ryanodine receptor (i.e., RyR^{R4496C} mutation) or binding to calsequestrin 2 (CASQ2). In SANC, Ca^{2+} signaling directly and indirectly mediates pacemaker function.

We address here the following research questions: (i) what coupled-clock mechanisms and pathways mediate pacemaker mutations associated with CPVT in basal and in response to β -adrenergic stimulation? (ii) Can different mechanisms lead to the same CPVT-related pacemaker pauses? (iii) Can the mutation-induced deteriorations in SANC function be reversed by drug intervention or gene manipulation?

We used a numerical model of mice SANC that includes membrane and intracellular mechanisms and their interconnected signaling pathways. In the basal state of RyR^{R4496C} SANC, the model predicted that the Na^+ - Ca^{2+} exchanger current (I_{NCX}) and T-type Ca^{2+} current (I_{CaT}) mediate between changes in Ca^{2+} signaling and SANC dysfunction. Under β -adrenergic stimulation, changes in cAMP-PKA signaling and the sodium currents (I_{Na}), in addition to I_{NCX} and I_{CaT} , mediate between changes in Ca^{2+} signaling and SANC automaticity pauses. Under basal conditions in $\text{Casq2}^{-/-}$, the same mechanisms drove changes in Ca^{2+} signaling and subsequent pacemaker dysfunction. However, SANC automaticity pauses in response to β -AR stimulation were mediated by I_{CaT} and I_{Na} . Taken together, distinct mechanisms can lead to CPVT-associated SANC automaticity pauses. In addition, we predict that specifically increasing SANC cAMP-PKA activity by either a pharmacological agent (IBMX, a phosphodiesterase (PDE) inhibitor), gene manipulation (overexpression of adenylyl cyclase 1/8) or direct manipulation of the SERCA phosphorylation target through changes in gene expression, compensate for the impairment in SANC automaticity. These findings suggest new insights for understanding CPVT and its therapeutic approach.

1. Introduction

Catecholaminergic polymorphic ventricular tachycardia (CPVT) is an inherited arrhythmia characterized by stress-induced ventricular arrhythmias, syncope, and/or sudden cardiac death [1–3]. CPVT patients are conventionally treated by β -blockers, but still some are at a high risk of arrhythmic events and require implantable cardiac defibrillators (ICD) [4,5].

Approximately 60% of identified mutations in CPVT patients

concern the gene encoding the cardiac ryanodine receptor (RyR2), the SR Ca^{2+} release channel [6–8]. In approximately 2% of the identified mutations, the gene encoding the SR Ca^{2+} buffer calsequestrin 2 (CASQ2), is affected [9,10], and much less frequently, mutations have been found in other proteins which can bind and modulate RyR2 function, such as triadin and calmodulin [11]. Thus, mishandling of intracellular Ca^{2+} is clearly the origin of CPVT.

Because Ca^{2+} mediates pacemaker function, it is important to investigate the function of the heart's primary pacemaker, the sinoatrial

Abbreviations: AC, Adenylyl cyclase; AP, Action potential; CASQ2, Calsequestrin 2; CL, Cycle length; CPVT, Catecholaminergic polymorphic ventricular tachycardia; ISO, Isoproterenol; SAN, Sinoatrial node; β -AR, β -adrenergic receptors

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node (SAN), and its role in CPVT patients. Indeed, even though CPVT originates in the ventricle, deterioration of the SAN has been documented in CPVT patients [12]. Neco et al. [8] noted that the RyR2^{R4496C} mutation was associated with a high occurrence of Ca²⁺ sparks and Ca²⁺ waves in the diastolic period under β -adrenergic stimulation (β -AR), which led to abnormal action potential (AP) generation, and thus to impaired SAN automaticity with pauses (which are associated with life threatening arrhythmia). Similarly, a slower heart rhythm and high diastolic Ca²⁺ leak was observed in SAN tissues isolated from RyR2^{R420Q} knock-in mice [13]. In their investigation of the effect of loss of CASQ2 (Casq2^{-/-}) on SAN function, Glukhov et al. [14] observed perturbations in intracellular Ca²⁺ cycling, which led to SAN automaticity pauses, particularly under β -AR stimulation. Although both the RyR2^{R4496C} mutation and loss of CASQ2 result in Ca²⁺ cycling dysregulation, and ultimately to SAN dysfunction, the Ca²⁺-sensitive pathway altering pacemaker function remains unknown.

Ca²⁺ cycling is one of the main mechanisms coupling molecules within the coupled-clock system, an ensemble of membrane currents and signaling pathways that ignite action potential in SAN cells (SANCs) [15]. Ca²⁺ can directly affect the function of membranal molecules and Ca²⁺-sensitive intracellular mechanisms. Indirect effects can be also obtained through either changes in membrane potential or Ca²⁺ activation of calmodulin that leads to activation of cAMP/PKA signaling [16]. In this regard, it is unknown if and how changes in cAMP/PKA signaling or other non-Ca²⁺ channels and exchangers are involved in CPVT-related pacemaker dysfunction. Because direct regulation of cAMP/PKA signaling by β -AR stimulation can trigger arrhythmia, the dynamics of cAMP/PKA signaling and other non-direct Ca²⁺-regulated pathways in CPVT-related mutant SANCs should be investigated.

Numerical modeling is a very powerful and sometimes the only feasible tool to explore these biophysical questions, offering a theoretical basis for experimental observations and innovative treatments (e.g., Behar and Yaniv [17]). Using a numerical model, we investigated the coupled-clock mechanisms and pathways involved in arrhythmogenic events associated with RyR2^{R4496C} mutation [8] and loss of CASQ2 [14], under basal and β -adrenergic stimulation conditions. The model was also applied to predict the reversibility of mutation-induced deteriorations in SANC function by drug intervention or gene manipulation.

2. Materials and methods

2.1. Existing experimental results

Neco et al. [8] investigated the effect of the RyR2^{R4496C} mutation on SAN and SANC activity. Experimental results showed a slight decrease in SAN automaticity in the basal state, and an impaired chronotropic response of the SAN accompanied by SAN automaticity pauses, upon β -AR stimulation using isoproterenol (ISO). Experimental results also revealed that ISO significantly decreased the Ca²⁺ transient amplitude, Sarcoplasmic reticulum (SR) Ca²⁺ load, and L-type Ca²⁺ (I_{CaL}) current.

When investigating Casq2^{-/-} in SAN, Glukhov et al. [14] observed perturbations in intracellular Ca²⁺ cycling, which led to SANC automaticity pauses in the basal state (in 36% of the cases) and under ISO treatment (in 74% of the cases). Under ISO treatment, these pauses were preceded by an increase in the concentration of intracellular diastolic Ca²⁺ and a decrease in the amplitude of the intracellular Ca²⁺ transient. Moreover, when exposed to ISO, the maximum action potential (AP) cycle length (CL) shortens significantly, but the minimum and average AP CL remained the same in Casq2^{-/-} as in wild type (WT) SANCs.

2.2. The numerical model

We used our previously published model of the mouse SANC [17].

This model was built upon the published model of Kharche et al. [18], Yaniv et al. [16] and Behar et al. [19]. The model describes the dynamics of cell membrane potential, ion currents, Ca²⁺ cycling and post-translational modifications during action potential.

2.3. Model strategy

We used the numerical model to investigate two types of changes in the coupled-clock system: (i) changes in RyR2^{R4496C} SANCs and (ii) changes under loss of CASQ2 induced by the CASQ2 knockout [14].

The RyR channel was simulated by a 4-state Markov model (one activated and three inactivated states), with both constant and Ca²⁺-dependent transition rates [20]. The RyR^{R4496C} mutation affects the open probability of the RyR channel [21], possibly by increasing the sensitivity of the channel to Ca²⁺ ions in the subspace [22]. Thus, we simulated the RyR2^{R4496C} mutation by changing the following coefficients (Eqs. (5)–(8)): (1) reduction of k_{om} from 0.06 ms⁻¹ to 0.045 ms⁻¹. k_{om} is the rate transition constant from open state to re-activated (closed) state; reducing it leads to enhanced activity of the RyR channel; (2) reduction of n_{Ca} from 2 to 1.9 in the absence of ISO and to 1.7 in the presence of ISO. n_{Ca} is the power of subspace Ca²⁺ in the transition equations (Fig. 1.A); its reduction leads to RyR hypersensitivity to subspace Ca²⁺, which is enhanced under β -AR stimulation. β -AR stimulation used 20 nM ISO, as in the experiments of Neco et al. [8]. The CASQ2 buffer, which serves to bind Ca²⁺ in the SR, was simulated by a transition equation of the fractional occupancy of CASQ2 by Ca²⁺. Casq2^{-/-} was simulated by the following changes: (1) reduction of the total concentration of the CASQ2 buffer (CQ_{tot}) in the junctional SR (JSR, Fig. 1.B) from 10 mM to 2.6 mM; (2) reduction of the Ca²⁺ association constant for CASQ2 (k_{fCQ}) from 0.534 to 0.3 in the presence of ISO.

2.4. Model equations

Parameters from the following model equations were modified in order to generate the simulations. Refer to the supplement for the variables and constants.

2.4.1. RyR2^{R4496C} mutation

$$j_{SRCrel} = k_s \cdot OO \cdot ([Ca^{2+}]_{JSR} - [Ca^{2+}]_{sub}) \quad (1)$$

$$k_{CaSR} = MaxSR - (MaxSR - MinSR) / (1 + (EC_{50SR} / [Ca^{2+}]_{JSR})^{HSR}) \quad (2)$$

$$k_{oSRCa} = k_{oCa} / k_{CaSR} \quad (3)$$

$$k_{iSRCa} = k_{iCa} \cdot k_{CaSR} \quad (4)$$

$$dR/dt = (k_{im} \cdot RI - k_{iSRCa} \cdot [Ca^{2+}]_{sub} \cdot R) - (k_{oSRCa} \cdot [Ca^{2+}]_{sub}^n \cdot R - k_{om} \cdot OO) \quad (5)$$

$$\begin{aligned} dOO/dt &= (k_{oSRCa} \cdot [Ca^{2+}]_{sub}^n \cdot R - k_{om} \cdot OO) - (k_{iSRCa} \cdot [Ca^{2+}]_{sub} \cdot OO - k_{im} \cdot S) \end{aligned} \quad (6)$$

$$dS/dt = (k_{iSRCa} \cdot [Ca^{2+}]_{sub} \cdot OO - k_{im} \cdot S) - (k_{om} \cdot S - k_{oSRCa} \cdot [Ca^{2+}]_{sub}^n \cdot RI) \quad (7)$$

$$dRI/dt = (k_{om} \cdot S - k_{oSRCa} \cdot [Ca^{2+}]_{sub}^n \cdot RI) - (k_{im} \cdot RI - k_{iSRCa} \cdot [Ca^{2+}]_{sub} \cdot R) \quad (8)$$

We used $k_{om} = 0.06 \text{ ms}^{-1}$, $n_{Ca} = 2$ for WT, $k_{om} = 0.045 \text{ ms}^{-1}$, $n_{Ca} = 1.9$ for RyR^{R4496C} cells under basal condition and $k_{om} = 0.045 \text{ ms}^{-1}$, $n_{Ca} = 1.7$ for RyR^{R4496C} cells exposed to 20 nM ISO.

2.4.2. Loss of CASQ2

$$df_{CQ} = k_{fCQ} \cdot [Ca^{2+}]_{JSR} \cdot (1 - f_{CQ}) - k_{bCQ} \cdot f_{CQ} \quad (9)$$

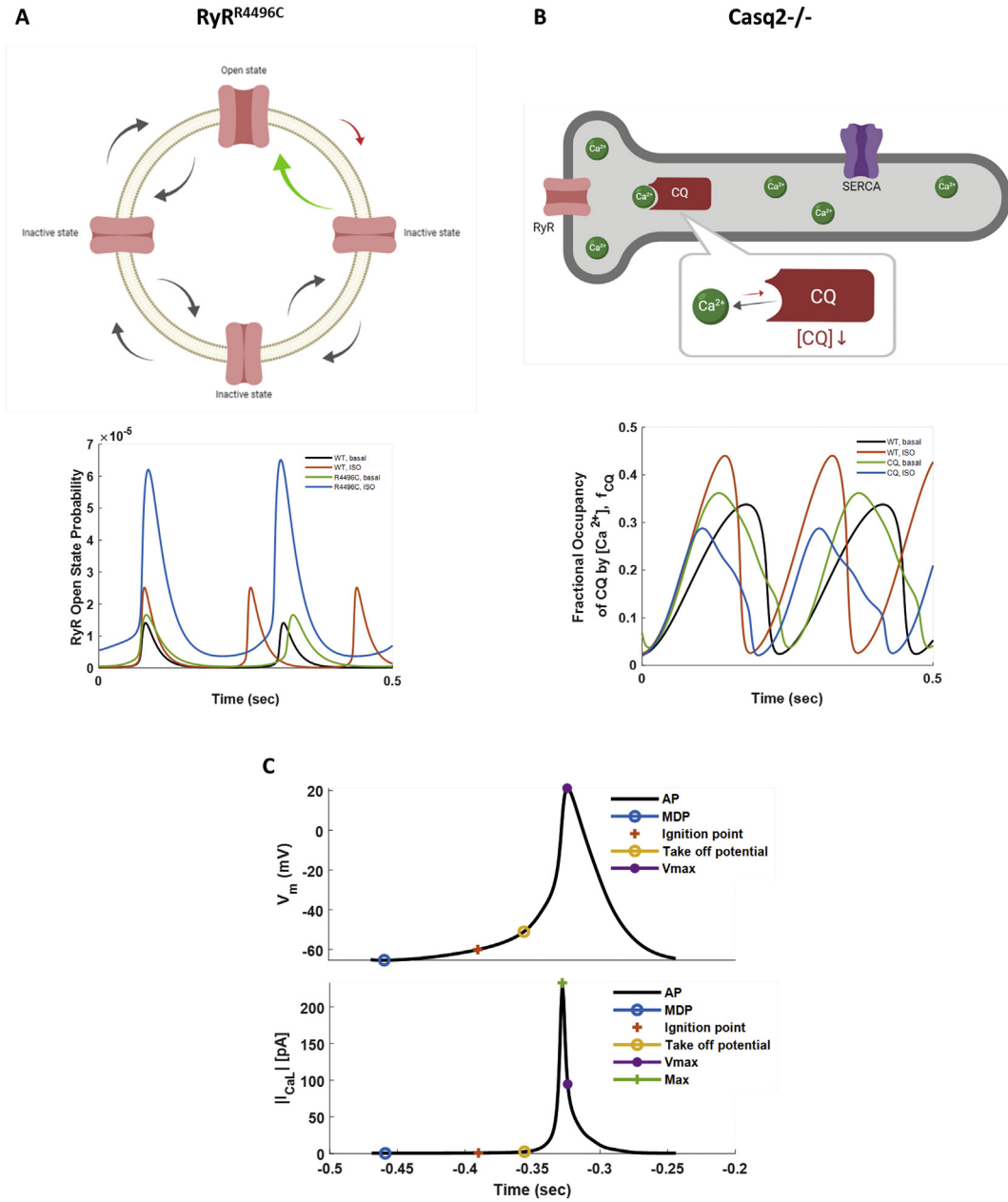


Fig. 1. Modeling of mutations affecting membrane and sarcoplasmic reticulum (SR) molecules. (A): Top: the RyR^{R4496C} four-state channel was modeled with a greater opening probability and hypersensitivity to Ca²⁺. Bottom: simulated RyR open-state probability in WT without (black) and with (red) isoproterenol, and RyR^{R4496C}, without (green) and with (blue) isoproterenol. (B): Top: the CASQ2 knockout was modeled by a reduced concentration of calsequestrin in the SR and reduced Ca²⁺ association constant. Bottom: simulated fractional occupancy of calsequestrin by Ca²⁺ in WT, without (black) and with (red) isoproterenol, and RyR^{R4496C}, without (green) and with (blue) isoproterenol. (C) Membrane potential and representative membrane current marked at maximal diastolic potential (MDP), ignition point, take-off potential, maximal voltage (Vmax) and maximal value (Max) points. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

$$d[Ca^{2+}]_{SR}/dt = j_{tr} - j_{SR\text{Carel}} - CQ_{tot} \cdot df_{CQ} \quad (10)$$

We used $CQ_{tot} = 10 \text{ mM}$, $k_{fCQ} = 0.534 \frac{1}{\text{mM} \cdot \text{ms}}$ for WT, $CQ_{tot} = 2.6 \text{ mM}$, $k_{fCQ} = 0.534 \frac{1}{\text{mM} \cdot \text{ms}}$ for Casq2^{-/-} SANC under basal condition and $CQ_{tot} = 2.6 \text{ mM}$, $k_{fCQ} = 0.3 \frac{1}{\text{mM} \cdot \text{ms}}$ for Casq2^{-/-} exposed to 10 nM ISO.

It was shown that SR Ca²⁺ levels is one of the regulators of SR Ca²⁺ release termination [23]. In our model, RyR release and CASQ2 concentration affect and are affected by the jSR Ca²⁺ level. Thus, RyR kinetics and CaSQ2 concentration also affect SR Ca²⁺ release termination.

The software was run in MATLAB (The MathWorks, Inc., Natick,

MA, USA). Numerical integration was performed using the MATLAB ode15s stiff solver, and the model simulations were run for 2000 s to ensure that steady state was reached. Computation was performed on an Intel(R) Core(TM) i7-4790 CPU @ 3.60GHz machine with 8 GB RAM. The source code of the numerical model is available at <http://bioelectric-bioenergetic-lab.net.technion.ac.il>.

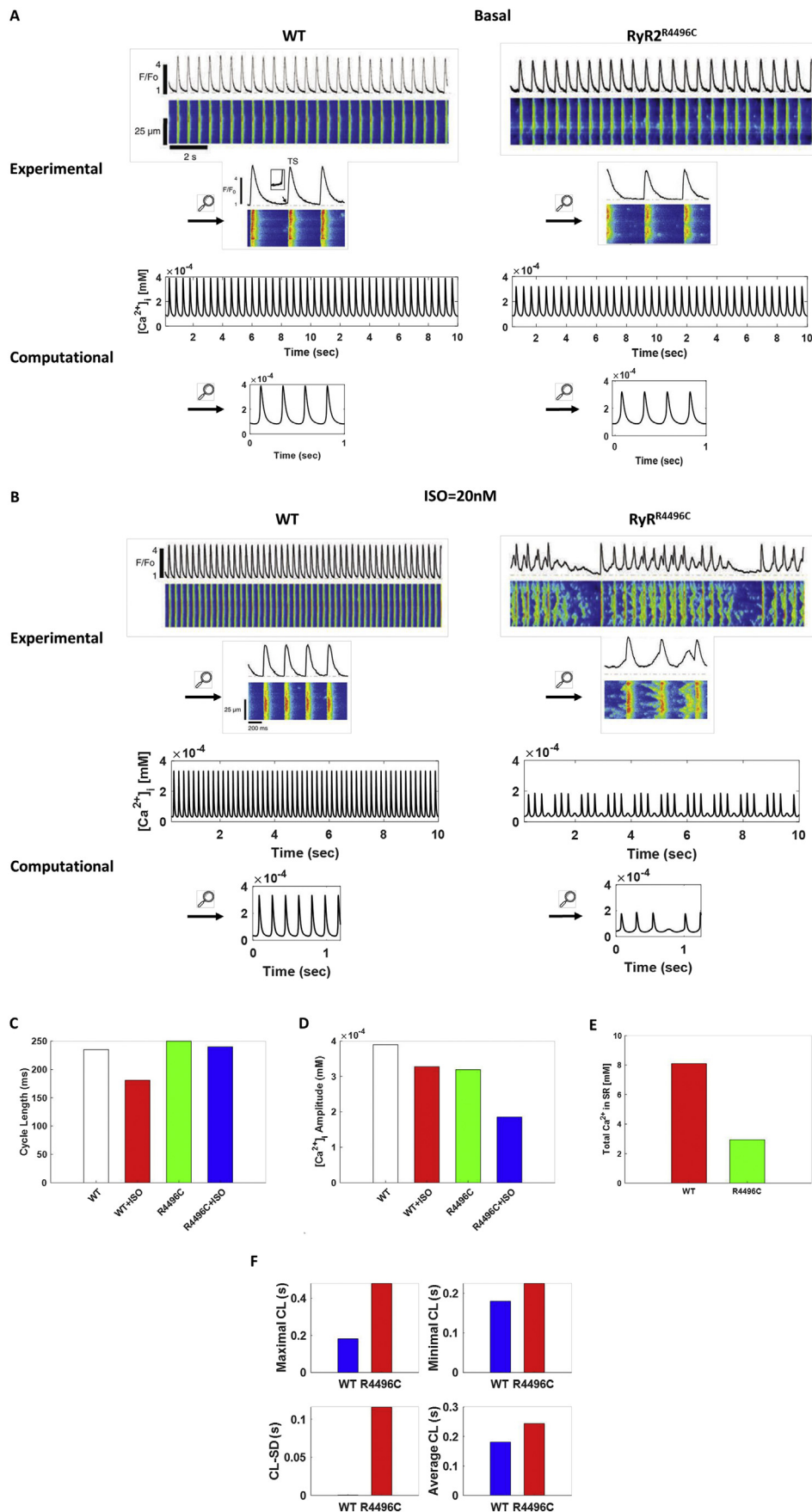


Fig. 2. Impaired Ca^{2+} handling and pauses in beating under β -adrenergic stimulation in $\text{RyR2}^{\text{R4496C}}$ sinoatrial node cells. (A) Top: Representative line-scan images and their fluorescence profile of Ca^{2+} transients recorded in WT (left) and $\text{RyR2}^{\text{R4496C}}$ (right) SANCs under basal conditions [8]. Bottom: Simulated amplitude of Ca^{2+} transient in WT (left) and $\text{RyR2}^{\text{R4496C}}$ (right) SANC models. (B) Top: Representative line-scan images and their fluorescence profile of Ca^{2+} transients recorded in WT (left) and $\text{RyR2}^{\text{R4496C}}$ (right) SANCs in response to 20 nM ISO [8]. Bottom: Simulated amplitude of Ca^{2+} transients in WT (left) and $\text{RyR2}^{\text{R4496C}}$ (right) SANC models. (C): Average cycle length of WT SANCs in the absence (white bar) and presence (red bar) of 20 nM ISO and of $\text{RyR2}^{\text{R4496C}}$ SANCs in the absence (green bar) and presence (blue bar) of 20 nM ISO, as calculated from the model simulation. (D): $[\text{Ca}^{2+}]_i$ maximal amplitude in WT SANCs in the absence (white bar) and presence (red bar) of 20 nM ISO and of $\text{RyR2}^{\text{R4496C}}$ SANCs in the absence (green bar) and presence (blue bar) of 20 nM ISO, as calculated from the model simulation. (E): Simulated bar graph of the maximal Ca^{2+} concentration in the SR of WT and $\text{RyR2}^{\text{R4496C}}$ SANCs exposed to ISO. (F) Action potential cycle length statistics of the $\text{RyR2}^{\text{R4496C}}$ model. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

3. Results

3.1. Pacemaker automaticity and Ca^{2+} cycling are impaired in $\text{RyR2}^{\text{R4496C}}$ SANCs

To verify reproducibility of the experimental results using our numerical model, we first simulated the AP and Ca^{2+} transient in WT and $\text{RyR2}^{\text{R4496C}}$ SANCs under basal conditions and compared them to existing experimental results. Note that the model simulates SANC at physiological temperature (37 °C), while experiments were performed at room temperature (22 °C). We therefore only refer to percent changes in parameters and not to their absolute values.

The model successfully reproduced Ca^{2+} activity in both WT and $\text{RyR2}^{\text{R4496C}}$ SANCs (Fig. 2.A). Note that the experiments reported by Neco et al. [8] were performed in SAN tissues and our model studies SANCs, therefore differences in the values of global parameters (such as cycle length) are expected. Under β -AR stimulation conditions (20 nM ISO), aberrant Ca^{2+} behavior with SANC automaticity pauses was observed in $\text{RyR2}^{\text{R4496C}}$ SANCs (Fig. 2.B), which aligned with the experimental data [8]. More specifically, under basal conditions, $\text{RyR2}^{\text{R4496C}}$ SANCs showed a slight increase (6%) in AP CL as compared to WT. We stipulated that any change < 10% is considered as a non-remarkable change because it cannot be measured experimentally. However, upon exposure to ISO, $\text{RyR2}^{\text{R4496C}}$ SANC exhibited an attenuated response in comparison to WT (4% decrease in AP CL in $\text{RyR2}^{\text{R4496C}}$ compared to 22% decrease in WT). Of note, to account for the pauses, the average AP CL of $\text{RyR2}^{\text{R4496C}}$ SANCs in response to ISO was calculated as the median AP CL (Fig. 2.C). In parallel, in WT SANCs, ISO slightly decreased the maximal $[\text{Ca}^{2+}]_i$ amplitude by 15% compared to the basal conditions. This reduction was remarkably enhanced (42%) in $\text{RyR2}^{\text{R4496C}}$ SANCs (Fig. 2.D) treated with ISO as compared to their control. Finally, the amount of Ca^{2+} stored in the SR was calculated as the cumulative sum of Ca^{2+} located in the JSR, in the network SR and in association with CASQ2. The model predicted a reduction of 63% in SR Ca^{2+} load in the $\text{RyR2}^{\text{R4496C}}$ SANC compared to WT under ISO stimulation (Fig. 2.E), which aligns with the experimental data [8].

3.2. Automaticity of $\text{Casq2}^{-/-}$ SANCs is preserved under β -adrenergic stimulation and Ca^{2+} cycling is impaired

To further verify the ability of our numerical model to reproduce the experimental results, we simulated the AP and Ca^{2+} transient in WT and $\text{Casq2}^{-/-}$ SANCs under basal conditions and compared the results to the experimental data. The average AP CL in $\text{Casq2}^{-/-}$ SANCs was similar to WT, as documented in the experimental results of Glukhov et al. [14]. Under β -AR stimulation conditions (10 nM ISO), the model generated aberrant Ca^{2+} behavior with SANC automaticity pauses in the $\text{Casq2}^{-/-}$ SANCs (Fig. 3.A), which was in accordance with the experimental data [14]. For $\text{Casq2}^{-/-}$ SANCs under β -AR stimulation, the model predicted no measurable differences in the minimum AP CL or in the average AP CL as compared to WT SANCs (Fig. 3.B). Namely, the AP CL outside of the pauses in $\text{Casq2}^{-/-}$ and WT SANCs under β -AR stimulation, is comparable. These predictions are in agreement with the experimental results.

3.3. Impaired coupled-clock mechanisms in $\text{RyR2}^{\text{R4496C}}$ SANCs lead to pacemaker dysfunction

Although stress-dependent changes in the Ca^{2+} transient of $\text{RyR2}^{\text{R4496C}}$ SANCs have been reported, no data exist on changes in other Ca^{2+} -related and -unrelated coupled-clock mechanisms. The numerical model can illustrate the direct and indirect interaction of other pacemaker functions arising from the coupled-clock effect. We analyzed changes in coupled-clock mechanisms using six different parameters: the maximal value of each component, the values of the component at

maximal diastolic potential (MDP), ignition point (IP, $\text{dV/dt} = 0.15 \text{ V/s}$ [24]), take-off potential (TOP, $\text{dV/dt} = 0.5 \text{ V/s}$ [24]), and maximal AP (V_{max}) and the area under the curve (AUC). Under basal conditions, no measurable differences in the AP CL, IP or TOP were found between WT cells and $\text{RyR2}^{\text{R4496C}}$ SANCs (Fig. 4.A). Due to changes in RyR2 kinetics, remarkable reductions in JSR Ca^{2+} (maximal value, values at MDP, IP, TOP, V_{max} and AUC, Fig. 4.B) and Ca^{2+} release from the SR (maximal value, value at V_{max} and AUC, Fig. 4.C) were documented. Consequently, Ca^{2+} in the subspace decreased (maximal value, value at V_{max} and AUC, Fig. 4.D). Note that at MDP, IP and TOP, there was an increase in Ca^{2+} release from the SR and in Ca^{2+} in the subspace due to the mutation effect. In contrast, the peak amplitude of cytosolic Ca^{2+} ($[\text{Ca}]_i$) and its value at V_{max} decreased only modestly (10–20%, Fig. 4.E). Regarding the membrane currents (Fig. 4.F–J and Supplementary Fig. S1), the sodium-calcium exchanger current (I_{NCX} , maximal value and value at V_{max} , Fig. 4.F) and T-type Ca^{2+} current (I_{CaT} , maximal value, Fig. 4.G) decreased. The amplitudes of the sodium current (I_{Na}), L-type Ca^{2+} current (I_{CaL}), funny current (I_f) and transient outward potassium current (I_{to}) decreased moderately (10–20%). Thus, I_{NCX} and I_{CaT} are the main membrane currents that mediate between SR Ca^{2+} cycling and pacemaker function in $\text{RyR2}^{\text{R4496C}}$ SANCs under basal conditions. Because $[\text{Ca}^{2+}]_i$ did not change, cAMP and PKA levels (that are activated by Ca^{2+} in SANCs) were comparable in $\text{RyR2}^{\text{R4496C}}$ and WT cells (Fig. 4.K–L).

Next, we explored the coupled-clock mechanisms in $\text{RyR2}^{\text{R4496C}}$ SANCs under β -AR stimulation, before and during SANC automaticity pauses (Fig. 5.A). There were measurable changes in IP and TOP values, together with an increase in ignition phase duration (i.e., the time between IP and TOP). Junctional SR Ca^{2+} levels (maximal value, values at MDP, IP, TOP, V_{max} and AUC) decreased remarkably, as did the Ca^{2+} flux from the SR (maximal value, value at V_{max} and AUC) and subspace Ca^{2+} (maximal value, value at V_{max} and AUC, Fig. 5.B–D), and consequently, intracellular Ca^{2+} (maximal value, value at V_{max} and AUC, Fig. 5.E). Regarding the membrane currents, I_{NCX} amplitude decreased remarkably (due to a decrease in subspace Ca^{2+}), as did I_{CaT} and I_{Na} amplitudes (Fig. 5.F–H). I_f (Fig. 5.I) and I_{to} (Fig. 5.J) amplitudes decreased, albeit to a lesser extent. Note that although I_{CaL} amplitude (Supplementary Fig. S2) decreased as well, the difference was comparable to that observed between WT and $\text{RyR2}^{\text{R4496C}}$ under basal conditions, and thus, likely does not contribute to the ISO-induced SANC automaticity pauses. In parallel, cAMP and PKA exhibited an attenuated response to β -AR stimulation (Fig. 5.K–L). In summary, through changes in cAMP/PKA signaling and I_{Na} , Ca^{2+} directly and indirectly leads to impaired $\text{RyR2}^{\text{R4496C}}$ SANC function in response to β -AR stimulation.

3.4. Impairment of coupled-clock mechanisms in $\text{Casq2}^{-/-}$ affects SANC function

Although stress-dependent changes in Ca^{2+} transients have been reported in $\text{Casq2}^{-/-}$, no data exist regarding changes in other coupled-clock mechanisms due to Ca^{2+} -related and -unrelated changes. As shown with $\text{RyR2}^{\text{R4496C}}$, the numerical model can illustrate the indirect effect of a knockout model on other pacemaker functions arising from the coupled-clock mechanism. Under basal conditions, no measurable difference was found between the AP CL, IP or TOP of $\text{Casq2}^{-/-}$ and WT SANCs (Fig. 6.A). Reduced CQ_{tot} in $\text{Casq2}^{-/-}$ SANCs did not result in considerable changes in JSR Ca^{2+} (Fig. 6.B), but led to remarkable reductions in Ca^{2+} flux from the SR (maximal value, value at V_{max} and AUC, Fig. 6.C) and in subspace Ca^{2+} (maximal value and value at V_{max} , Fig. 6.D). Only a moderate change in intracellular Ca^{2+} (maximal value and value at V_{max} , Fig. 6.E) was detected. Consequently, there were no measurable differences in cAMP and PKA levels between $\text{Casq2}^{-/-}$ and WT SANCs. Regarding membrane currents, a remarkable decrease in I_{NCX} (50% decrease in maximal value and 87% at V_{max} , Fig. 6.F) and I_{CaT} (64% decrease in maximal value, Fig. 6.G) was documented in

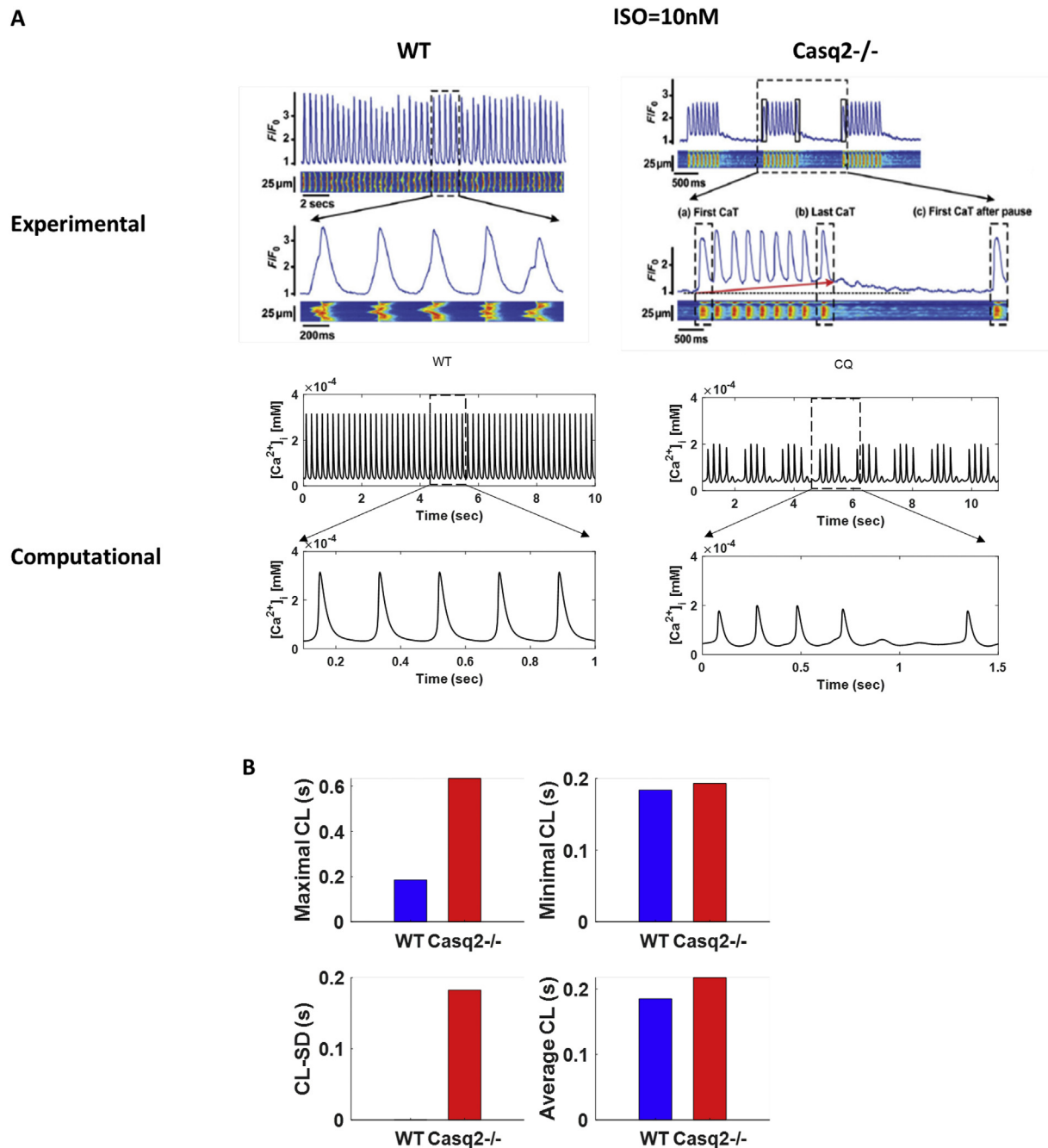


Fig. 3. Sinoatrial node cell automaticity pauses observed under β -adrenergic stimulation in Casq2^{-/-} cells. **A.** Top: Representative line-scan images and their fluorescence profile of Ca²⁺ transients recorded in WT (left) and Casq2^{-/-} (right) SANCs in response to 10 nM ISO [8]. Bottom: Simulated amplitude of Ca²⁺ transients in WT (left) and Casq2^{-/-} (right) SANC models. **B.** Action potential cycle length statistics of the Casq2^{-/-} model.

parallel to a remarkable but smaller decrease in amplitude of I_{Na} (23%, Fig. 6H), I_{CaL} (25%, Fig. 6I), I_{to} (35%, Fig. 6J) and I_f (21%, Fig. 6K), and sustained depolarization-induced outward current (I_{sus} , 23%, Fig. 6L). Taken together, as shown for RyR2^{R4496C}, I_{NCX} together with I_{CaT} are the main membrane currents (compared to other currents, see Fig. 6 and Supplementary Fig. S3) that mediate between changes in Ca²⁺ cycling and pacemaker function in Casq2^{-/-} SANCs under basal conditions.

Next, we explored the coupled-clock mechanisms in Casq2^{-/-} SANCs under β -AR stimulation with SANC automaticity pauses (Fig. 7A). IP did not change. However, TOP was found to be measurably different as compared to WT. The reduced CQ_{tot} in Casq2^{-/-} SANCs did not lead to a remarkable change in the peak JSR Ca²⁺

measured in WT SANCs (Fig. 7B). However, because of the shift in TOP there is a reduction in JSR Ca²⁺ at that point with increase at MDP due to the mutation effect. Remarkable reduction in Ca²⁺ flux from the SR (maximal value, value at V_{max} and AUC, Fig. 7C), subspace Ca²⁺ (maximal value and value at V_{max} , Fig. 7D) and intracellular Ca²⁺ (maximal value and value at V_{max} , Fig. 7E) were documented. Note that during the pauses, there was an abrupt decrease in Ca²⁺ in both SR compartments. Because cAMP and PKA dynamics do not act on a beat-to-beat basis and because intracellular Ca²⁺ was not altered during the ignition phase between the pauses, phosphorylation signaling, i.e., cAMP and PKA levels, did not change remarkably as compared to WT SANCs. Regarding membrane currents (Fig. 7F–L), a remarkable decrease in I_{CaT} amplitude (Fig. 7G) and I_{Na} amplitude (Fig. 7H) was

evident. Decreases were also evident in peak I_{NCX} and I_f (Fig. 7.F and K); however, their magnitudes were similar to the basal state and thus, these currents seemingly do not contribute to the pauses. Interestingly, the reduction in peak I_{CaL} (Fig. 7.I) and I_{sus} (Fig. 7.L) were abolished under β -AR stimulation. In summary, under β -AR stimulation, I_{CaT} together with I_{Na} are the main membrane currents (Fig. 7 and Supplementary Fig. S4) that mediate between changes in Ca^{2+} signaling and pacemaker function in $Casq2^{-/-}$ SANCs.

3.5. Therapeutic pathways compensate for the effects of coupled-clock mutations

We then used our computational model to explore whether the effects of mutation-associated pacemaker dysfunction on AP CL and SANC automaticity pauses under ISO can be reversed by drug perturbations or gene manipulation that directly target the SAN (e.g., injected directly into the SAN area or targeting of specific SAN genes such HCN4). To this end, we tested the effect of the phosphodiesterase (PDE) inhibitor IBMX, direct manipulation of a SERCA phosphorylation target (i.e., phospholamban) and overexpression of adenylyl cyclase (AC) 1 and/or 8. SAN has a high PDE activity [25]. Thus, in order to restore the beating rate, we explored the effect of the nonspecific PDE inhibitor, IBMX. Note that similar approaches have been shown to restore aged dysfunctional SAN [17]. A concentration of 3 μ M IBMX shortened the AP CL of $RyR2^{R4496C}$ SANCs to that of WT SANCs (Supplementary Fig. S5.A). SERCA has been shown to be the most potent molecule that can increase or decrease the SAN beating rate [19] and changes in its function can be therapeutic [25]. Thus, we explored whether increasing SERCA activity can restore SAN dysfunction. Increasing the SERCA P_{up} (pumping rate) from 0.02 mM/ms to 0.035 mM/ms (via gene manipulation), shortened the AP CL of $RyR2^{R4496C}$ SANCs to that of WT SANCs (Supplementary Fig. S5.B). Thus, phosphorylation-related pharmacological interventions or gene manipulation can reverse $RyR2^{R4496C}$ mutant-associated pacemaker dysfunction. SAN expresses AC 1 and 8 which are Ca^{2+} -activated ACs [26,27], leading to high phosphorylation activity in the SAN [16] even without β -AR. Increasing the presence of AC 1 and 8 (by increasing K_{AC} from 0.0735 to 0.2 min^{-1}) decreased the AP CL of $RyR2^{R4496C}$ SANCs to that of WT SANCs (Supplementary Fig. S5.C), as recently suggested [28]. Note that similar approaches have been shown to restore aged dysfunctional SAN [17,29]. These three drug perturbation/gene manipulations also eliminated the SANC automaticity pauses under β -AR stimulation.

Next, we used the same interventions to explore the behavior of $Casq2^{-/-}$ SANCs. As average AP CL was similar for $Casq2^{-/-}$ and WT SANCs under basal conditions, we only explored whether SANC automaticity pauses were eliminated under β -AR stimulation and the above interventions. As seen in $RyR2^{R4496C}$, IBMX (3 μ M) or upregulation of AC 1 and 8 (by increasing K_{AC} to 0.2 min^{-1}) eliminated the pauses. However, when the SERCA phosphorylation target was manipulated through changes in gene expression, the pauses were not eliminated, even at a low ISO concentration (10 nM). (See Fig. 8.)

4. Discussion

In this paper we explored internal mechanisms and interactions of non-measurable parameters in CPVT SANC model.

Our first major conclusion is the identification of CPVT mutation-associated clock mechanisms and pathways that mediate the deterioration of pacemaker function. Here, we investigated two CPVT-associated mutations affecting $RyR2$ and $CASQ2$. In RyR^{R4496C} , the mutation directly causes hypersensitivity of the RyR channel to Ca^{2+} , resulting in enhanced release of Ca^{2+} from the SR during diastole (Fig. 4.C), which mainly affected I_{NCX} and I_{CaT} (Fig. 4.F–G). The changes in I_{NCX} were mediated through changes in subspace Ca^{2+} , and I_{CaT} was affected by the changes in the diastolic membrane potential. Both currents affected the membrane potential and led to small but

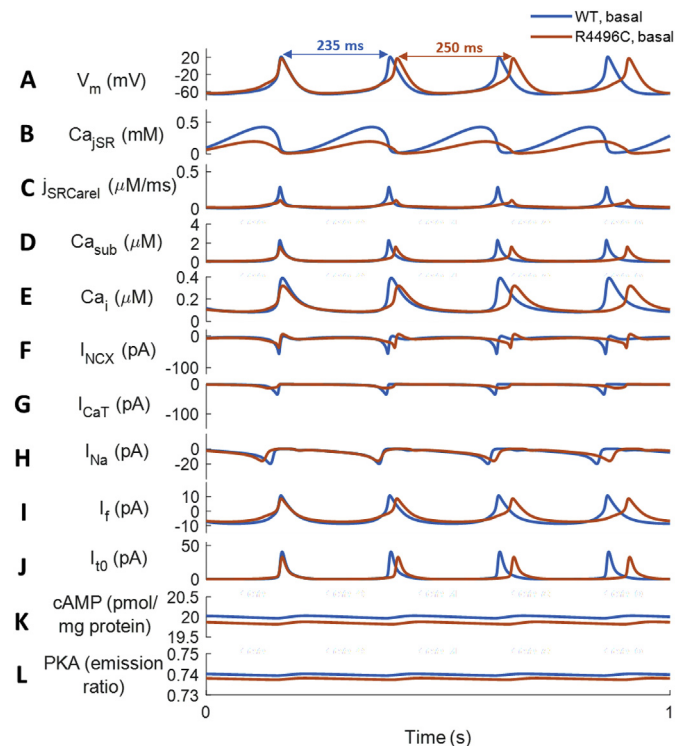


Fig. 4. Major currents and Ca^{2+} cycling in simulated basal $RyR2^{R4496C}$ sinoatrial node cells. Coupled-clock function of WT (blue) and $RyR2^{R4496C}$ (red) SANCs modeled under basal conditions: (A) membrane voltage (V_m) and representative cycle lengths, (B) Ca^{2+} concentration in the junctional SR compartment (Ca_{JSR}), (C) flux of Ca^{2+} exiting the SR ($j_{SRCaRel}$), (D) Ca^{2+} concentration in the subspace (Ca_{sub}), (E) intracellular Ca^{2+} concentration (Ca_i), (F) Na^+-Ca^{2+} exchanger current (I_{NCX}), (G) T-type Ca^{2+} current (I_{CaT}), (H) sodium current (I_{Na}), (I) HCN (funny) current (I_f), (J) transient potassium current (I_{to}), (K) cyclic adenosine monophosphate level (cAMP) and (L) protein kinase A level (PKA). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

measurable changes in I_{to} , I_f and I_{Na} . We therefore conclude that I_{NCX} and I_{CaT} , which are remarkable contributors to early depolarization, are the main membrane currents that lead to loss in pacemaker function due to the changes in Ca^{2+} cycling in $RyR2^{R4496C}$ SANCs. Note that because only small changes were predicted in cytosolic (Fig. 4.E) or SR (Fig. 4.B) Ca^{2+} in mutant SANCs compared to WT, under basal conditions, the SANCs continue to operate normally with the same AP CL and without pauses. For $Casq2^{-/-}$ cells, the model showed that under basal conditions, the AP CL in SANCs did not differ remarkably from that of WT SANCs (Fig. 6.A). This result was in agreement with the experimental results [14]. However, the numerical model made it possible to investigate, for the first time, the mechanisms causing $Casq2^{-/-}$ SANCs dysfunction. Reduced ability to buffer Ca^{2+} in the SR led to a reduced release of Ca^{2+} during diastole (Fig. 6.C) and to a decreased peak of subspace Ca^{2+} (Fig. 6.D). As in the $RyR2^{R4496C}$ mutation, I_{NCX} and I_{CaT} (Fig. 6.F–G) were the main membrane currents compensating for the loss in pacemaker function, due to the loss of $CASQ2$, in this case. However, because in $Casq2^{-/-}$ cells the reduction in subspace Ca^{2+} is more substantial than in $RyR2^{R4496C}$ cells, the reduction in I_{NCX} was greater. Moreover, the reduction in subspace Ca^{2+} also led to a small but measurable reduction in I_{CaL} in $Casq2^{-/-}$ cells compared to $RyR2^{R4496C}$ cells. Note that a reduction in I_{CaL} was documented in $RyR2^{R4496C}$ cells in response to ISO [8]. However, our model predicted only a 10% reduction. It is possible that the mutation in RyR drove a reduction in the number of L-type channels (see limitations section). Here, we only modeled a reduction in I_{CaL} , because of the reduction in subspace Ca^{2+} and changes in the membrane potential. Moreover,

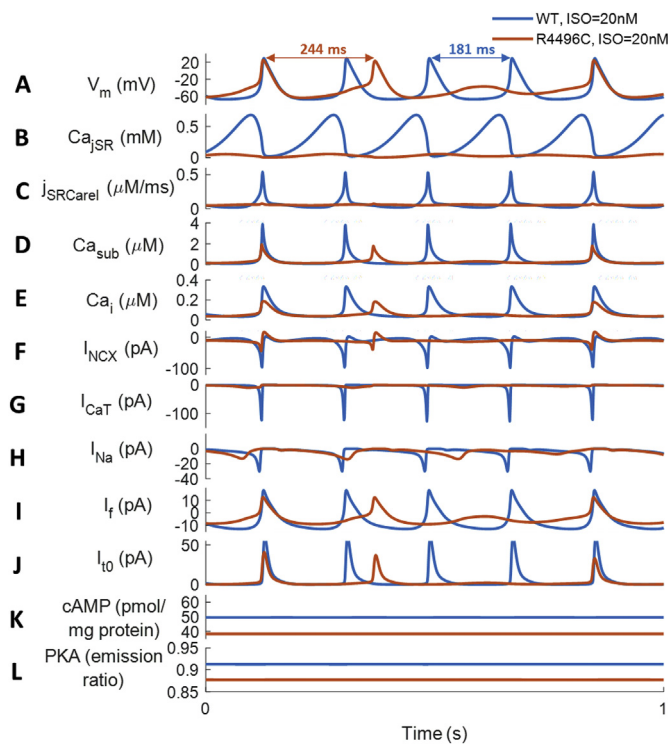


Fig. 5. Major currents and Ca^{2+} cycling in simulated $\text{RyR2}^{\text{R4496C}}$ sinoatrial node cells under β -adrenergic stimulation. Coupled-clock function of WT (blue) and $\text{RyR2}^{\text{R4496C}}$ (red) SANCs subjected to 20 nM ISO: (A) membrane voltage (V_m) and representative cycle lengths, (B) Ca^{2+} concentration in the junctional SR compartment (Ca_{JSR}), (C) flux of Ca^{2+} exiting the SR (j_{SRcarel}), (D) Ca^{2+} concentration in the subspace (Ca_{sub}), (E) intracellular Ca^{2+} concentration (Ca_i), (F) Na^+ - Ca^{2+} exchanger current (I_{NCX}), (G) T-type Ca^{2+} current (I_{CaT}), (H) sodium current (I_{Na}), (I) HCN (funny) current (I_f), (J) transient potassium current (I_{to}), (K) cyclic adenosine monophosphate level (cAMP) and (L) protein kinase A level (PKA). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

because a Ca^{2+} -induced inactivation mechanism exists, and a reduction in subspace Ca^{2+} was simulated, it is possible that this mechanism inhibits further reduction in I_{CaL} .

Our second major conclusion is that different intracellular mechanisms can lead to SANC automaticity pauses in response to stress simulated by β -AR stimulation. In the case of the $\text{RyR2}^{\text{R4496C}}$ mutation, the hyperactivity of the RyR channel during diastole was enhanced, which led not only to reduced systolic release of Ca^{2+} from the SR, but also to reduced cytosolic and SR Ca^{2+} (Fig. 4.B–D). These results are in accordance with the experimental results of Neco et al. [8], who showed that the SR Ca^{2+} load was remarkably reduced with the presence of ISO. Our model suggests that under these conditions, it is not I_{NCX} alone that leads to the loss in pacemaker function due to changes in Ca^{2+} cycling, but also I_{CaT} and I_{Na} (Fig. 4.F–H). While I_{NCX} was reduced because of reduced subspace Ca^{2+} , changes in I_{CaT} and I_{Na} were mediated by changes in the membrane potential during the diastolic phase. These three currents led not only to an attenuated response to ISO, but also to SANC automaticity pauses (Fig. 5). In the $\text{Casq2}^{-/-}$ model, I_{CaT} and I_{Na} , but not I_{NCX} , currents contributed to the formation of pauses. The main reason is because there was no further reduction in subspace Ca^{2+} compared to the basal state. Interestingly, although both the I_{CaT} and I_{Na} currents contributed to the formation of pauses in the $\text{Casq2}^{-/-}$ model, they did not affect the average AP CL in comparison to their effect in $\text{RyR2}^{\text{R4496C}}$ SANCs. We hypothesize that this effect may be due to a minimal contribution of I_{NCX} in the $\text{Casq2}^{-/-}$ model. A comparison of the two CPVT mutations shows that SANC automaticity pauses can be generated with different AP patterns resulting from the

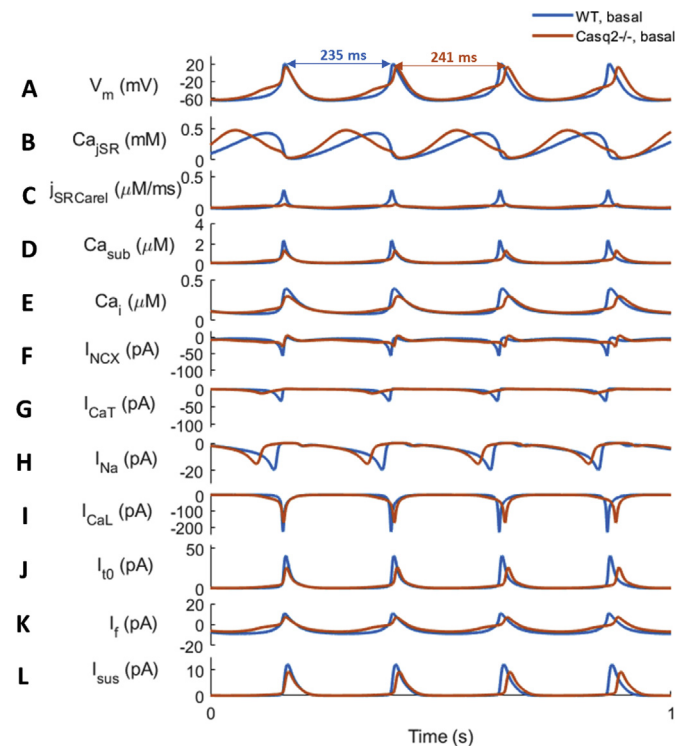


Fig. 6. Major currents and Ca^{2+} cycling in simulated basal $\text{Casq2}^{-/-}$ sinoatrial node cells. Coupled-clock function of WT (blue) and $\text{Casq2}^{-/-}$ (red) SANCs under basal conditions: (A) membrane voltage (V_m) and representative cycle lengths, (B) Ca^{2+} concentration in the junctional SR compartment (Ca_{JSR}), (C) flux of Ca^{2+} exiting the SR (j_{SRcarel}), (D) Ca^{2+} concentration in the subspace (Ca_{sub}), (E) intracellular Ca^{2+} concentration (Ca_i), (F) Na^+ - Ca^{2+} exchanger current (I_{NCX}), (G) T-type Ca^{2+} current (I_{CaT}), (H) sodium current (I_{Na}), (I) L-type Ca^{2+} current (I_{CaL}), (J) transient potassium current (I_{to}), (K) HCN (funny) current (I_f) and (L) sustained potassium current (I_{sus}). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

alteration of different mechanisms. Thus, CPVT-associated mutations cannot be conclusively identified just by measuring the average AP CL. Both CPVT mutations were associated with SANC automaticity pauses in response to stress simulated by β -AR stimulation. Thus, based on the simulation results, a reduction in I_{Na} below a certain level reduced the stability of the system. A question arises as to why SANC automaticity pauses occur only in response to stress (simulated by β -AR stimulation), in both models. In $\text{RyR2}^{\text{R4496C}}$ SANCs, pauses only occur when the decrease in I_{NCX} reaches a certain level, and is accompanied by a reduction in I_f and I_{Na} . These changes only occur at higher cAMP/PKA signaling levels (for I_f), reduction in intracellular Ca^{2+} (for I_{NCX}) and changes in the ignition phase (for I_{Na} activation) provoked by β -AR stimulation. In $\text{Casq2}^{-/-}$ SANCs, pauses occur only when the decrease in I_{NCX} reaches a certain level and is accompanied by a reduction in I_{Na} . Similarly to the $\text{RyR2}^{\text{R4496C}}$ model, these changes only occur upon reductions in intracellular Ca^{2+} (for I_{NCX}) and changes in the ignition phase (for I_{Na} activation) provoked by β -AR stimulation.

Our third major conclusion is that changes in phosphorylation signaling pathway contribute to the deterioration of pacemaker function associated with SANC mutations. In SANCs, the cAMP/PKA signaling pathway is activated by Ca^{2+} -calmodulin. Thus, changes in intracellular Ca^{2+} can lead to changes in cAMP/PKA activity. While in $\text{RyR2}^{\text{R4496C}}$ SANCs the model predicted a reduction in cAMP and PKA levels under β -AR stimulation compared to WT, there was no measurable difference in these signaling levels in $\text{Casq2}^{-/-}$ and WT SANCs when ISO concentrations were low (compared to high concentration, Supplementary Figs. S6–S7). These results can be explained by a

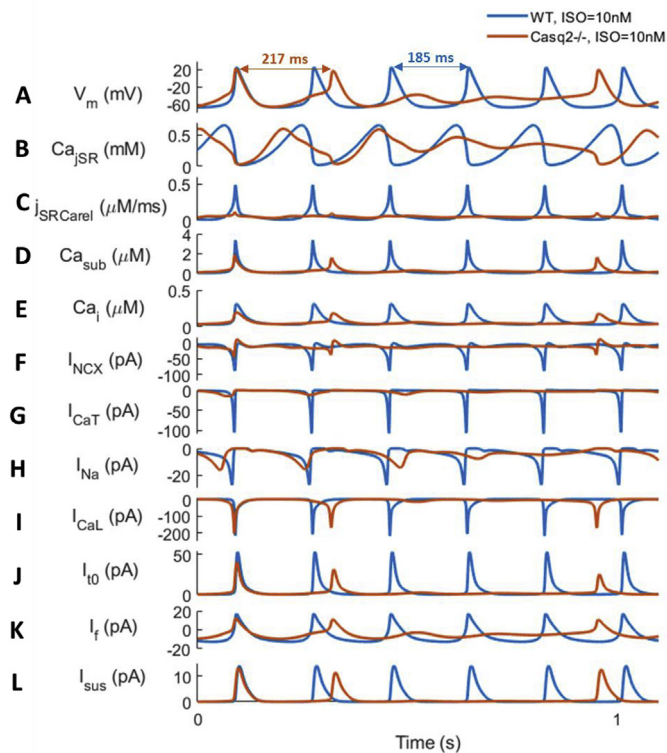


Fig. 7. Major currents and Ca^{2+} cycling in simulated $\text{Casq2}^{-/-}$ sinoatrial node cells under β -adrenergic stimulation. Coupled-clock function of WT (blue) and $\text{Casq2}^{-/-}$ (red) SANCs subjected to 10 nM ISO: (A) membrane voltage (V_m) and representative cycle lengths, (B) Ca^{2+} concentration in the junctional SR compartment ($\text{Ca}_{J,SR}$), (C) flux of Ca^{2+} exiting the SR ($j_{SR,Calc}$), (D) Ca^{2+} concentration in the subspace (Ca_{Sub}), (E) intracellular Ca^{2+} concentration (Ca_I), (F) the Na^+ - Ca^{2+} exchanger current (I_{NCX}), (G) T-type Ca^{2+} current (I_{CaT}), (H) sodium current (I_{Na}), (I) L-type Ca^{2+} current (I_{CaL}), (J) transient potassium current (I_{T0}), (K) HCN (funny) current (I_f) and (L) sustained potassium current (I_{sus}). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

reduction in peak intracellular Ca^{2+} in RyR2^{R4496C} SANCs and no measurable changes in intracellular Ca^{2+} in $\text{Casq2}^{-/-}$ SANCs. Note, however, that because cAMP/PKA signaling does not change on a beat-to-beat basis, it cannot affect or be affected by short SANC automaticity pauses that impact intracellular Ca^{2+} . Finally, the model suggests that the dynamics of cAMP/PKA can contribute, in some cases, to the deterioration of mutant SANC function and are therefore important to monitor.

Our fourth major conclusion is that indirect drug perturbation and gene expression can partially reverse CPVT-associated pacemaker dysfunction (Supplementary Fig. S5). The rationale of our idea of drug/gene expression therapy is rooted in two facts: (i) the drug will be delivered directly to the SAN area and will not directly affect the ventricular function. Similarly, gene expression will be localized in the SAN area; (ii) the SAN has a unique AC system; in contrast to Ca^{2+} -deactivated ACs in ventricles, SAN has AC 1 and 8, which are Ca^{2+} -activated ACs [26,27]. Moreover, SAN has a high PDE activity [25]. Thus, changes in AC-cAMP/PKA signaling is highly potent. Use of either a PDE inhibitor or increasing Ca^{2+} activation of AC, restored the AP CL of RyR2^{R4496C} SANCs to the WT level and stopped SANC automaticity pauses in both CPVT models. Note that the level of IBMX used (3 μM) was in physiological range [30]. These results strengthen the conclusion that a change in even one molecule of the coupled-clock system can affect the other molecules as well as affect the AP CL [31]. Note that although manipulation of the SERCA phosphorylation target in RyR2^{R4496C} SANCs eliminated SANC automaticity pauses, it failed to do

so in $\text{Casq2}^{-/-}$ SANCs, possibly due to the smaller influence of Ca^{2+} SR in the $\text{Casq2}^{-/-}$ mutation relative to its influence on the other model types. Importantly, two interventions eliminated SANC automaticity pauses in both CPVT models. These conclusions have important clinical implications because these pauses are the source of life-threatening arrhythmia. Thus, the presented model provided not only insights into the mechanisms mediating CPVT-related pacemaker dysfunction, but also identified therapeutic targets. Future experiments are needed to validate the model predictions. Note that Zhao et al. [32] showed that AC5 overexpression under certain conditions produces arrhythmias similar to CPVT. However, we suggest here to increase AC 1/8 and not AC 5/6 and to also overexpress AC1/8 only in SAN tissue.

Analysis of changes in coupled-clock components using six time-dependent parameters provided some interesting insights: (i) At IP, I_{NCX} was higher in both mutations as compared to WT, because of the increase in diastolic Ca^{2+} . (ii) Because of its kinetics, I_{CaT} is meaningless at IP and TOP, so its analysis should only be performed at its maximal value and at V_{max} . (iii) When ISO was applied, I_{NCX} changed its profile from baseline and thus it is hard to compare its value at other points then its maximal value. (iv) While some of the currents remarkably changed as compared to WT, their magnitude was at least one order below that of other currents, and thus were assumed to have a negligible effect.

4.1. Limitations

Limitations of this work included the simulation of global and not local Ca^{2+} releases. Thus, it cannot differentiate spark duration, length and amplitude, rather, only the ensemble of all these parameters. As a result, it cannot generate a related effect, such as early diastolic depolarization. In addition, while the model predicted no pauses in mutant SANCs under basal conditions, they were observed in 36% of $\text{Casq2}^{-/-}$ SANCs [14]. This can be partially explained by the deterministic nature of the model, which does not include stochastic behavior of the channels. Future development of such a model will shed light on the underlying mechanisms at work in CPVT and other related mutations. A stochastic model would also be valuable for investigation of the mechanisms underlying changes in local Ca^{2+} release dynamics and the appearance of Ca^{2+} waves documented in RyR2^{R4496C} SANCs under β -AR stimulation [8]. The experimental model of mutant mice was also limited in that, although a specific gene may be knocked out or knocked in, other genes may compensate for the altered function, as discussed in the Nature Neuroscience editorial [33]. Because none of the experimental studies [8,14,34] investigated the effect of the mutations on the expression of all membrane and SR proteins, this could not be taken into account in the numerical model. Because some experiments were performed at room temperature, the AP firing rate was normalized, but may have been inexact due to non-linearities and differences in temperature dependence (Q10) of the various channels/transporters. The model explored only the PKA-related signaling and not CaMKII signaling, in SAN dysfunction. Further data related to CaMKII activity in the SAN will be necessary to explore its effects on pacemaker function. Finally, because the model only simulates single SAN cell function, it cannot provide insights regarding the function of the entire heart.

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jmcc.2020.04.017>.

Author contributions

Y.Y., J.B. and L.A.G conceived and designed the research; L.A.G and J.B. performed simulations and analyzed the simulated data; J.B., Y.Y and L.A.G drafted the manuscript; L.A.G and J.B. prepared the figures; L.A.G, J.B. A.M.G and Y.Y edited and revised the manuscript, and approved the final version.

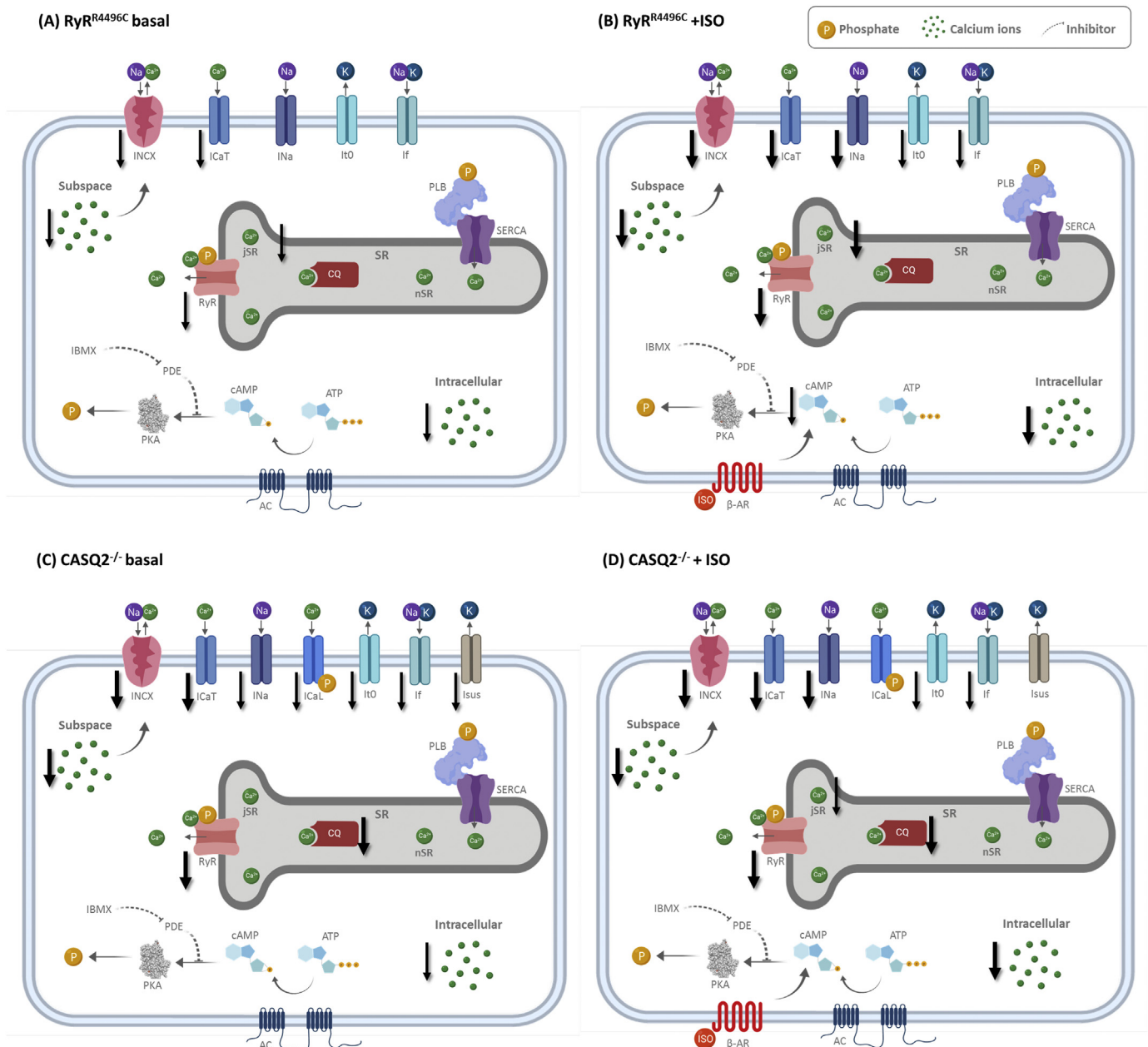


Fig. 8. Schematic illustration of the main mechanisms involved in CPVT-related sinoatrial node dysfunction. (A) RyR^{R4496C} SANC model under basal conditions; (B) RyR^{R4496C} SANC model with 20 nM ISO; (C) Casq2^{-/-} SANC model under basal conditions and (D) Casq2^{-/-} SANC model with 10 nM ISO. The thinner and thicker arrows represent a 20–40% change and over 40% change as compared to WT, respectively.

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